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Thermal and dynamic modelling of permanent-magnet synchronous motors (PMSM) driven electric vehicle powertrain with dual-loop control strategy

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This paper presents a comprehensive electro-thermal model of a Permanent Magnet Synchronous Motor (PMSM)-driven electric vehicle (EV) powertrain integrated with a dual-loop control strategy for dynamic speed and current regulation. The model was developed in MATLAB/Simulink for speed tracking and torque response under varying load conditions. Thermal ports corresponding to the stator windings (HA, HB, HC) and rotor core (HR) were activated to capture heat transfer behaviour during operation. Simulation results demonstrated that the proposed control system effectively maintained the reference speed of 3,500 rpm across all torque levels, with rapid settling time (~0.4 s). An observed trend indicated that higher load torque resulted in faster convergence to the reference speed, accompanied by higher current demand and increased thermal stress, most notably within the rotor core. When sequential load torques were applied up to 120 Nm, temperatures exceeded safe limits, leading to thermal runaway. To address this, a three-stage proportional cooling strategy was implemented, successfully limiting temperature rise below 120 °C and restoring steady-state operation without derating. The results validate the importance of integrated electro-thermal modelling and staged cooling control in improving the reliability, safety, and performance of PMSM-based EV powertrains.

KEYWORDS

dual-loop control, electric vehicles (EVs), electro-thermal simulation, field-oriented control (FOC), permanent magnet synchronous motor (PMSM), staged cooling, thermal modelling

1 Introduction

The transition towards sustainable and energy-efficient transportation has significantly increased the adoption of electric vehicles (EVs) (Guo et al., 2023; T et al., 2023). At the core of these vehicles lies the electric powertrain, which directly influences performance, reliability, and energy efficiency (Dasari et al., 2025; Mohammed et al., 2025). Among the various electric motor technologies, permanent-magnet synchronous motors (PMSM) have emerged as the dominant choice for EV applications owing to their

high torque density, fast dynamic response, and superior efficiency (Hegde et al., 2025; Monadi et al., 2024; Azom and Khan, 2025). However, to fully leverage these advantages, robust control strategies and thermal management must be integrated into the motor-drive system.

The real-world operation of EVs involves frequent speed variations and load disturbances, making it essential to adopt advanced control mechanisms that can maintain stability and performance under dynamic conditions (Bhalla et al., 2025). A dual-loop control strategy comprising an outer-loop speed controller and an inner-loop current controller is widely recognised for its ability to ensure accurate torque and speed regulation in PMSM drives (Jiang and Yang, 2025). In particular, Field-Oriented Control (FOC) has gained prominence for decoupling the motor's flux and torque-producing components, thereby enabling independent control of stator currents (Yousef and Abdelmaksoud, 2015).

Beyond electromechanical control, thermal behaviour is a critical aspect that directly affects motor longevity and performance. Temperature rises in motor windings and core components due to resistive and core losses can lead to efficiency degradation, insulation failure, and permanent magnet demagnetisation. Therefore, thermal modelling of the PMSM, including stator winding (HA, HB, HC) and rotor (HR) heat paths, is crucial for assessing the operational safety and designing effective cooling strategies.

This study presents a comprehensive thermal and dynamic modelling framework for a PMSM-driven EV powertrain under realistic driving conditions. A dual-loop control system is implemented, where the output of the speed controller is fed into a current controller structured using the FOC methodology. The PMSM is modelled using the Simscape environment with its thermal ports enabled to simulate heat generation and propagation through HA, HB, HC, and HR. The simulation was carried out under different and varying load torque conditions to emulate practical driving scenarios with thermal stress. The results offer valuable insights into both the control effectiveness and the thermal limitations of PMSM drives in electric vehicles.

The remainder of this study is organised as follows. Section 2 reviews the relevant literature on PMSM-based electric powertrains, dual-loop control strategies, and thermal modelling approaches for EV applications. Section 3 details the methodology, including the motor parameter specifications, controller design, thermal port modelling, and simulation setup in the Simscape environment. Section 4 presents and discusses the simulation results, focusing on the motor performance, current response, speed tracking, and thermal behaviour under different load conditions. Section 5 concludes the study with key findings and implications for EV powertrain design and suggests directions for future research.

2 Literature review

The advancement of electric vehicle (EV) powertrains has driven extensive research into optimizing motor control, thermal management, and simulation methodologies. The Permanent Magnet Synchronous Motor (PMSM) has become central to these developments due to its superior efficiency, torque density, and controllability. However, realizing its full potential in EV applications requires not only precise electromechanical control but

also robust thermal regulation, especially under varying operational loads. This section reviews existing literature in five interconnected domains relevant to this study: (i) the role and suitability of PMSM in EV propulsion, (ii) dual-loop control strategies including speed and current control with Field-Oriented Control (FOC), (iii) thermal modelling approaches for accurately predicting and managing heat generation in PMSM, (iv) simulation tools and environments used for integrated electromechanical and thermal modelling, and (v) a synthesis of research gaps that this present study aims to address.

2.1 PMSM in electric vehicle applications

The Permanent Magnet Synchronous Motor (PMSM) has emerged as a preferred solution in electric vehicle (EV) propulsion systems due to its high efficiency, power density, and torque-per-ampere characteristics (Liao et al., 2022). PMSMs outperform induction and DC motors in performance-critical EV applications, especially at low and medium power ranges (Ali Khan, 2024; Mohammad and Jaber, 2022). Comparative studies Mohammad and Jaber (2022), and Sakunthala et al. (2017) have highlighted the PMSM's superior suitability for EVs over BLDC and induction motors due to its compact design and precise controllability. Additionally, the adoption of PMSM in in-wheel applications has prompted further study into the design trade-offs and compact thermal management strategies for space-constrained environments (Karnavas et al., 2016).

2.2 Dual-loop control strategies for PMSM drives

A dual-loop control strategy combining an outer-loop speed controller and an inner-loop current controller is widely implemented in PMSM-based systems to ensure high dynamic performance. Field-Oriented Control (FOC) remains the dominant technique, allowing decoupled control of torque and flux via d-q axis transformations (Ali Khan, 2024; Ghanayem et al., 2023; Ramesh et al., 2022). Comparative analysis of tuning methods, including manual, mathematical, and autotuning strategies, demonstrate that tuning significantly impacts the responsiveness and stability of the motor control system (Fayyaz et al., 2024). Moreover, robust alternatives such as predictive by Murali et al. (2021), fuzzy logic by Feng et al. (2017), and adaptive PR controllers by Ghanayem et al. (2023) have been proposed to reduce torque ripples and enhance speed tracking under load disturbances.

2.3 Thermal modelling of PMSM for EV applications

As the PMSM operates under high torque demands in EVs, thermal modelling becomes essential to prevent performance degradation due to excessive winding or rotor temperatures. Several studies have applied Lumped Parameter Thermal Networks (LPTNs) and Finite Element Methods (FEM) to simulate temperature rise and validate heat dissipation designs (Liang et al., 2024; Abubakar et al., 2022; Khaledidoost et al., 2022). Research shows that coupling electrical and thermal models increases simulation fidelity, particularly in compact or high-speed motors (Karnavas et al., 2016;

Sanghai and Burade, 2024). The temperature profiles of stator windings (HA, HB, HC) and rotor (HR) ports are particularly critical, requiring accurate modelling of conduction, convection, and potential cooling strategies such as air or liquid flow (Dan et al., 2023; Murugan et al., 2025).

2.4 Modelling and simulation tools in PMSM-EV research

MATLAB/Simulink and Simscape environments are widely used in PMSM simulation due to their flexibility and block-level abstraction that facilitates system integration. Researchers have demonstrated the design of customizable EV powertrains, integrating electrical and thermal subsystems for detailed performance analysis under realistic drive cycles (Zhang et al., 2022; Tomar et al., 2021). High-fidelity simulations allow visualization of current, speed, and thermal behaviour under transient load conditions and are vital in validating new control strategies (Murali et al., 2021; Le, 2024). Despite the computational load, such simulations are increasingly viable with modern hardware and model simplification strategies like reduced-order modelling.

2.5 Summary of existing studies and identified gaps

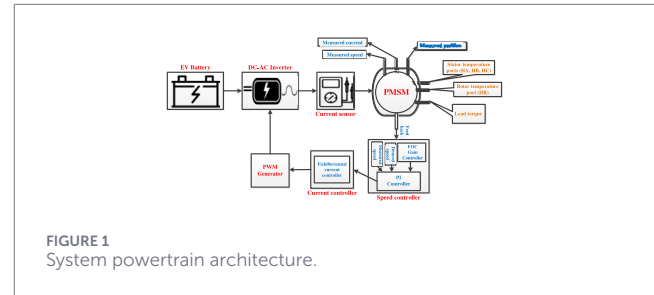
While existing literature provides extensive coverage of PMSM control and thermal modelling separately, there is limited integration of both domains in a unified simulation for real-time vehicle applications. Many studies focus either on torque ripple minimization (Feng et al., 2017) or on thermal characteristics (Liang et al., 2024; Abubakar et al., 2022; Sanghai and Burade, 2024) without dynamically linking them to dual-loop control performance. Additionally, the interaction between thermal stress and control loop stability under short-time, high-load torque scenarios remains underexplored. This study aims to bridge this gap by providing a co-simulation of thermal and dynamic performance in a PMSM-driven EV powertrain using FOC and dual-loop control under various load conditions.

3 Methodology

This section outlines the modelling framework, system configuration, control design, and thermal integration for the simulation of a PMSM-based electric vehicle (EV) powertrain. The model is developed using MATLAB/Simulink and Simscape, enabling co-simulation of electrical, mechanical, and thermal domains under realistic driving conditions. The simulation focuses on evaluating the dynamic response and thermal behaviour of the motor under variable load torque.

3.1 System architecture

The powertrain model shown in Figure 1 consists of the following primary components: PMSM Motor, Speed controller (PI-based) in the outer control loop, Current controller (Field-Oriented Control) in the inner loop, EV battery for the input DC voltage, PWM Generator, and Inverter to generate a controlled three-phase



excitation for the motor. The PMSM is modelled using Simscape Electrical's "Permanent Magnet Synchronous Motor" block with the thermal port enabled to capture heat generation and flow during operation.

3.2 Control strategy

A dual-loop control system is implemented to regulate the dynamic performance of the PMSM. This consists of:

- an outer-loop speed controller that generates a torque reference based on speed error, and
- an inner-loop current controller that regulates stator currents in the d-q reference frame using Field-Oriented Control (FOC)

3.2.1 Outer loop (speed control)

The speed control loop uses a Proportional-Integral (PI) controller to track the reference speed w_{ref} and generate the torque command T_e^* . The control law is defined in Equation 1 as:

$$T_e^*(t) = K_{p,w} e_w(t) + K_{i,w} \int e_w(t) dt \quad (1)$$

Where:

- T_e^* = commanded electromagnetic torque,
- $e_w(t) = w_{ref}(t) - w_{measured}(t)$ = speed error,
- $K_{p,w}, K_{i,w}$ = proportional and integral gains of the speed controller.

This torque reference is then passed to the current control loop for execution.

3.2.2 Inner loop (current control)

The inner loop performs torque control by regulating the d-q axis currents, which are obtained from the three-phase stator currents using Clarke and Park transformations (Chattopadhyay et al., 2011):

Clarke Transformation ($ABC \rightarrow \alpha\beta$) Equation 2:

$$\begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} = \frac{2}{3} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} \quad (2)$$

TABLE 1 PI controller gains.

Controller	(K_p)	(K_i)
Speed loop	0.8	50
d-axis current	0.5	200
q-axis current	0.5	200

Park Transformation ($\alpha\beta \rightarrow dq$) Equation 3:

$$\begin{bmatrix} i_d \\ i_q \end{bmatrix} = \begin{bmatrix} \cos(\theta_e) & \sin(\theta_e) \\ -\sin(\theta_e) & \cos(\theta_e) \end{bmatrix} \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} \quad (3)$$

Where:

i_d, i_q are the direct and quadrature currents,

θ_e is the electrical rotor angle obtained from the motor.

The control objectives are:

Set $i_d^* = 0$ to maximize torque per ampere and reduce losses.

Calculate i_q^* from the torque reference using the torque equation in Equation 4:

$$T_e = \frac{3}{2}p(\lambda i_q + (L_d - L_q)i_d i_q) \quad (4)$$

For surface PMSMs, $L_d \approx L_q$, so the torque simplifies to Equation 5:

$$T_e \approx \frac{3}{2}p\lambda i_q \Rightarrow i_q^* = \frac{2T_e}{3p\lambda} \quad (5)$$

Where: P = number of pole pairs, λ = permanent magnet flux linkage.

PI Current Controllers: Each axis (d and q) has its own PI controller expressed in Equations 6, 7.

$$v_d^* = K_{p,d}e_d + K_{i,d} \int e_d dt \quad (6)$$

$$v_q^* = K_{p,q}e_q + K_{i,q} \int e_q dt \quad (7)$$

Where $e_d = i_d^* - i_d$ and

$e_q = i_q^* - i_q$ are the current errors.

The voltage references v_d^* and v_q^* are transformed back to the three-phase voltages using inverse Park and inverse Clarke transformations, and used to generate PWM signals for the inverter.

The PI controllers used in the speed and current control loops were tuned empirically to achieve fast dynamic response with minimal overshoot. The controller gains are summarized in Table 1.

The tuning process involved iterative adjustment to achieve settling time < 0.5 s, minimal steady-state error, and stable thermal response.

3.2.3 PWM and inverter switching

The voltage commands are modulated using a sinusoidal PWM technique with a switching frequency of 10 kHz. The gating signals are then applied to a three-phase inverter, which energizes the motor windings accordingly. This is essential as PMSM motors require controlled current waveforms (electronic commutation) that are synchronized with the rotor position. A standard three-phase AC supply cannot provide this precision.

TABLE 2 PMSM model parameters (Courtay and Kelly, 2018; El-Refaie, 2013; Nikam et al., 2012).

Parameter	Value	Unit
Number of poles	8	–
Max power (Pmax)	35,000	W
Max torque (Tmax)	200	Nm
Stator resistance (Rs)	0.010087	Ω
d-axis inductance (Ld)	243.68e-6	H
q-axis inductance (Lq)	297.58e-6	H
Leakage inductance (L0)	121.84e-6	H
Flux linkage (FluxPM)	0.0440	Wb
Rated current	425	A
Rated speed	1,000	rpm
Rotor inertia (J)	0.1243	kg·m ²
DC link capacitance	1e-3	F
Rated voltage	325	V
PWM switching Freq	10,000	Hz

3.2.4 Simscape blocks used

The following blocks from Simscape Electrical are employed:

- FOC Default Controller Gains: Predefined tuned PI gains for current control.
- Field-Oriented Current Controller: Implements transformations, control logic, and PWM signal generation.
- PMSM with thermal port: Receives the control voltages and generates electromechanical outputs and thermal losses.

These blocks are preconfigured to support vector control through dynamic decoupling of d - q currents, enabling accurate torque production and fast response.

3.3 PMSM model parameters

The PMSM is modelled with realistic, vehicle-grade specifications as shown in Table 1. The PMSM model used in this study adopts parameters representative of EV-grade permanent magnet machines specifications as shown in Table 2. These specification is consistent with prior works such as (Courtay and Kelly, 2018; El-Refaie, 2013; Nikam et al., 2012). These values reflect typical PMSM configurations used in high-performance EV applications and allow for accurate simulation of both torque and thermal dynamics.

3.4 Thermal modelling

The thermal behaviour of the PMSM is modelled using Simscape's thermal ports: HA, HB, HC (stator windings) and HR (rotor). These ports simulate internal heat generation and heat transfer between motor components and the ambient environment.

Each thermal port is connected to:

- i. Thermal Mass Blocks to model heat storage (thermal capacitance),
- ii. Conductive Heat Transfer Blocks for internal conduction between winding phases and rotor,
- iii. Convective Heat Transfer Blocks to simulate cooling (e.g., airflow or casing dissipation).

The thermal model employed in this study is based on a lumped parameter thermal network (LPTN), where the PMSM is represented by discrete thermal nodes corresponding to stator windings (HA, HB, HC) and rotor core (HR). This approach was selected due to:

- i. Its computational efficiency, enabling real-time electro-thermal co-simulation (Fu et al., 2025);
- ii. Its suitability for system-level EV powertrain studies (Chen, 2020); and
- iii. Its widespread adoption in PMSM thermal analysis literature (Wallscheid and Böcker, 2015).

3.4.1 Governing equations

Each thermal node follows the first-order differential thermal energy balance provided in Equation 8:

$$C_{th} \frac{dT}{dt} = Q_{gen} - Q_{loss} \quad (8)$$

Where: C_{th} = thermal capacitance of the component (J/°C), T = temperature (°C), Q_{loss} = heat lost by conduction/convection (W), Q_{total} = total heat generated within the PMSM and is expressed as given in Equations 9–11.

$$Q_{total} = Q_{cu} + Q_{core} + Q_{sw} \quad (9)$$

$$Q_{cu} = I^2 R_s \quad (10)$$

$$Q_{core} = P_{hysteresis} + P_{eddy} \quad (11)$$

$$Q_{sw} = f_{sw} \cdot E_{sw}$$

Where:

Q_c = Copper losses, Q_s = Core losses.

Q_{sw} = Switching losses.

Core losses were modelled using empirical parameters derived from manufacturer data, while switching losses were approximated based on inverter switching frequency.

Convective heat loss (to ambient) is shown in Equation 12

$$Q_{loss} = h \cdot A \cdot (T - T_{ambient}) \quad (12)$$

Where: h = convective heat transfer coefficient (W/m²·°C), A = exposed surface area of the winding or rotor (m²), $T_{ambient}$ = ambient temperature (°C).

3.4.2 Temperature control and protection logic

To emulate real-world electric vehicle (EV) safety systems, a three-stage cooling mechanism is implemented through variation of the convective heat transfer coefficient h . The system continuously

TABLE 3 Cooling activation logic.

Temperature range	Action taken
$T < T_{safe}$	No cooling needed
$T_{safe} \leq T < T_1$	Activate stage 1, (natural convection cooling)
$T_1 \leq T < T_2$	Activate stage 2, (forced air cooling)
$T_2 \leq T < T_{max}$	Activate stage 3 (liquid cooling)
$T \geq T_{max}$	Initiate motor derating or shutdown

monitors the temperature of each motor thermal port, HA, HB, HC (windings) and HR (rotor core) and dynamically engages stages of cooling as temperature rises as shown in Table 3.

Instead of abrupt switching, the cooling system is divided into three stages, activated based on threshold zones as shown in Table 3.

Where:

T_{safe} is Nominal safe temperature (e.g., 80 °C)

T_1 and T_2 are Intermediate thresholds (e.g., 100 °C, 115 °C)

T_{max} is Critical limit (e.g., 130 °C)

Let $T(t)$ be the port temperature, $Q_{cool}(t)$ can therefore be expressed as shown in Equation 13.

$$Q_{cool}(t) = \begin{cases} 0, & T(t) < T_{safe} \\ h_1 A (T - T_{amb}), & T_{safe} \leq T(t) < T_1 \\ h_2 A (T - T_{amb}), & T_1 \leq T(t) < T_2 \\ h_3 A (T - T_{amb}), & T_2 \leq T(t) < T_{max} \\ Derating/shutdown, & T(t) > T_{max} \end{cases} \quad (13)$$

Where:

$h_1 < h_2 < h_3$ are heat transfer coefficients representing increasing cooling intensities.

A is heat transfer area.

T_{amb} is ambient temperature.

These are modelled in Simscape using variable convective heat transfer coefficients and MATLAB Function blocks to control additional cooling branches and implement tiered logic and derating.

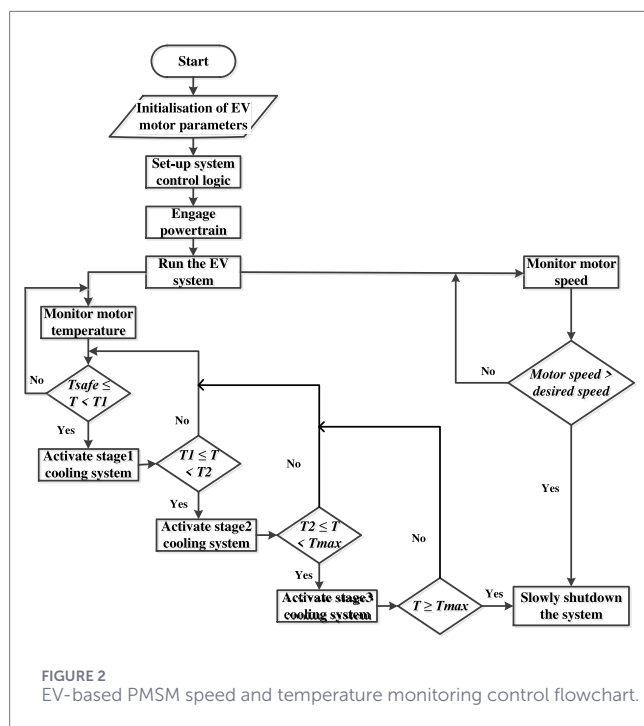
If the motor temperature exceeds T_{max} , and Stage 3 cooling is active, the system performs:

- i. Soft derating i.e., reduce torque command proportionally.
- ii. Hard shutdown i.e., open inverter switches (if temperature continues to rise), and halting motor activity to protect windings and rotor.

The goal here is to monitor and manage the temperature rise at each winding and rotor core. Proper thermal boundary conditions are applied to ensure that port temperatures stay within operational safety limits.

3.5 Load torque and simulation conditions

To emulate a real-world driving scenario, variable load torques are applied in intervals within 60-s simulation period, i.e., 0–10s (No load: 0 Nm), 10–20s, i.e., (Low load: 10 Nm), 20–30s, i.e., (Medium load: 50 Nm), 30–40s, i.e., (High load: 100 Nm), 40–50s, i.e., (Very high load: 150 Nm), 50–60s, i.e., Regenerative braking (No load: 0



Nm). The simulation is set for discrete solver with a step size of $1e-6$ s, ensuring numerical stability and time accuracy needed for high-frequency switching and transient thermal analysis.

The following outputs are observed and analysed: Three-phase stator currents (I_a , I_b , I_c), Rotor speed, Electromagnetic torque, and Temperatures at HA, HB, HC, and HR ports. All signals are displayed using scopes and logged for post-simulation analysis. Figure 2 shows the flowchart for EV-based PMSM speed and temperature monitoring control.

4 Results and discussion

This section presents and analyses the simulation results obtained from the PMSM-driven electric vehicle (EV) powertrain model incorporating dual-loop control and integrated thermal behaviour. The performance is evaluated in terms of dynamic response (speed, torque, and current) and thermal performance (temperature rise at HA, HB, HC, and HR ports). Two simulation durations were used: 10 s for dynamic performance analysis (speed and current response), and 60 s for thermal analysis due to slower heat dynamics.

4.1 Model overview

Figure 3 shows the Simulink model developed for thermal and dynamic simulation of a PMSM-based EV powertrain with dual-loop control. The outer loop regulates speed, while the inner loop controls the d-q currents via FOC. The PMSM block includes four thermal ports (HA, HB, HC, HR) connected to conductive and convective heat transfer blocks. A multi-stage cooling control algorithm was integrated to limit port temperatures to safe operating thresholds, as described in Subsection 3.4.2.

4.2 Constant torque simulations

Figure 4–8 illustrate the PMSM performance under constant load torques of 0 Nm, 10 Nm, 50 Nm, 100 Nm, and 150 Nm applied for 10 s each.

The results show that the dual-loop control strategy effectively maintained the desired reference speed of 3,500 rpm for all operating conditions. The outer-loop speed controller and the inner-loop current controller cooperatively ensured that any transient deviation caused by a change in torque demand was immediately compensated, resulting in stable and uniform steady-state speed across all tests.

A distinct trend observed in the simulations is that the higher the applied load torque, the faster the PMSM reached the reference speed. This behaviour stems from the control model's design, which prioritizes rapid speed convergence by increasing the torque command and, consequently, the stator current during acceleration. At higher torques, the controller injects a more i_q component, producing larger electromagnetic torque that accelerates the rotor more rapidly. This intentional compromise in control tuning ensures that speed regulation takes precedence over efficiency during start-up and load transitions, a technique also adopted in advanced field-oriented control implementations for traction drives (El-Refaie, 2013). However, this behaviour is dependent on controller tuning and not a generalized behaviour across all PMSM systems.

The current response follows this pattern. At motor start-up, a high inrush current was observed across all cases due to the need to overcome initial inertia. As the motor approached the reference speed, current gradually decreased and stabilized at a steady-state value dependent on the applied load torque. For instance, the steady current magnitude under 150 Nm was the highest, while for no-load conditions (0 Nm) was the lowest, confirming that current demand scales proportionally with mechanical load.

Regarding thermal performance, the temperature rises in the stator windings (HA, HB, and HC) remained almost identical, indicating uniform phase loading. However, the rotor thermal port (HR) exhibited a faster and higher temperature increase compared to the stator windings for all torque conditions. This is attributed to concentrated iron losses and reduced convective cooling efficiency within the rotor core, consistent with experimental observations in high-torque PMSM drives (El-Refaie, 2013).

For the highest torque case (150 Nm), HR reached its maximum temperature within a few seconds, while HA, HB, and HC exhibited slower but steady temperature growth. These results confirm that the developed dual-loop control strategy not only sustains the reference speed under varying mechanical loads but also captures thermal coupling behaviour between electrical and mechanical domains, validating the model's dynamic-thermal integration.

4.3 Variable torque simulation

Figure 9 shows the applied torque profile for the variable load test. The performance of the PMSM under a time-varying load torque profile is shown in Figure 10. The applied torque pattern

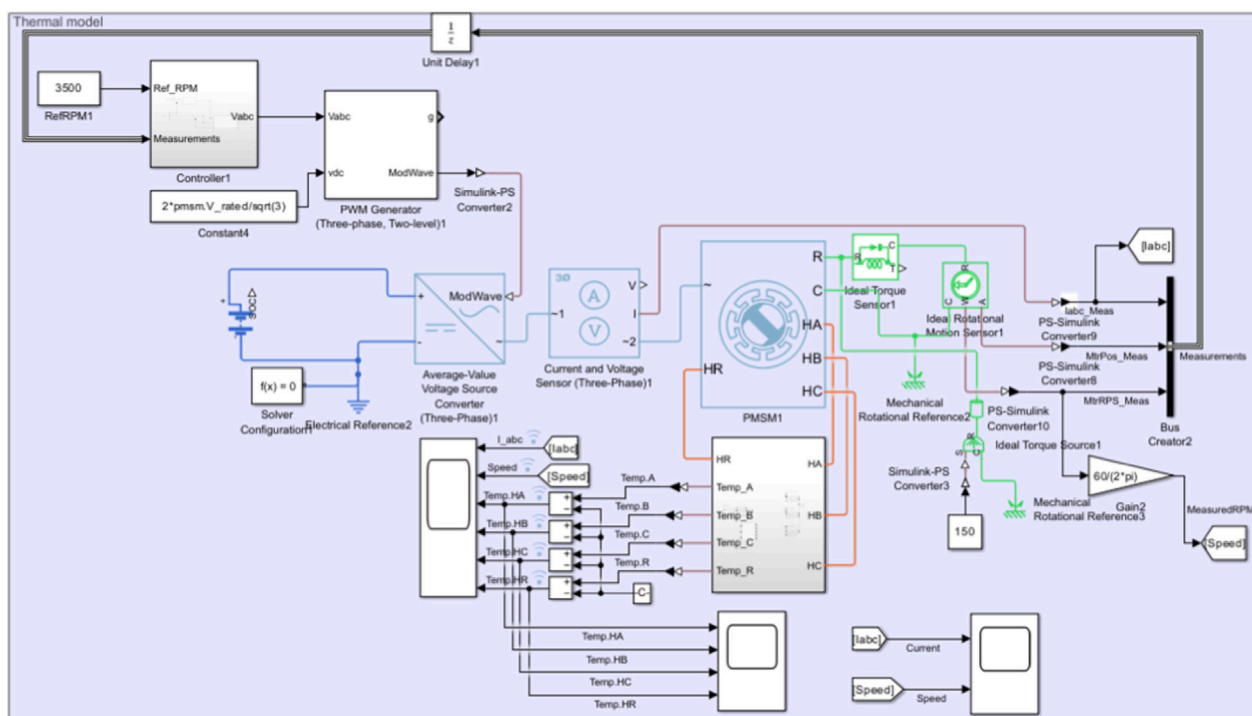


FIGURE 3
Simulink model of PMSM thermal and dynamic operation with dual-loop control strategy.

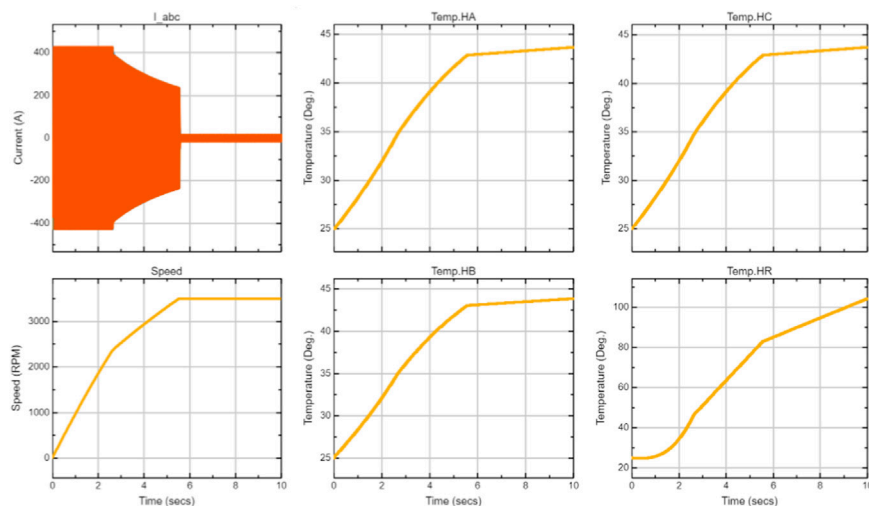


FIGURE 4
PMSM current, speed, and thermal port temperatures (HA, HB, HC, HR) under no-load torque (0 Nm) for 10 s.

followed six successive stages: 0 Nm (0–10 s), 10 Nm (10–20 s), 50 Nm (20–30 s), 100 Nm (30–40 s), 120 Nm (40–50 s), and a return to 0 Nm (50–60 s).

This test represents a realistic driving scenario involving acceleration, cruising, heavy-load climbing, and deceleration. During the initial no-load interval (0–10 s), the motor achieved its reference speed of 3,500 rpm rapidly and maintained it with a

minimal steady-state current, indicating efficient speed control at light load.

When the 10 Nm torque was applied at 10 s, both the current and temperature in the stator and rotor windings began to rise gradually, signifying increased electromagnetic loading and corresponding copper and iron losses. As torque successively increased in 10 s intervals, from 10 Nm up to 120 Nm, the current magnitude grew

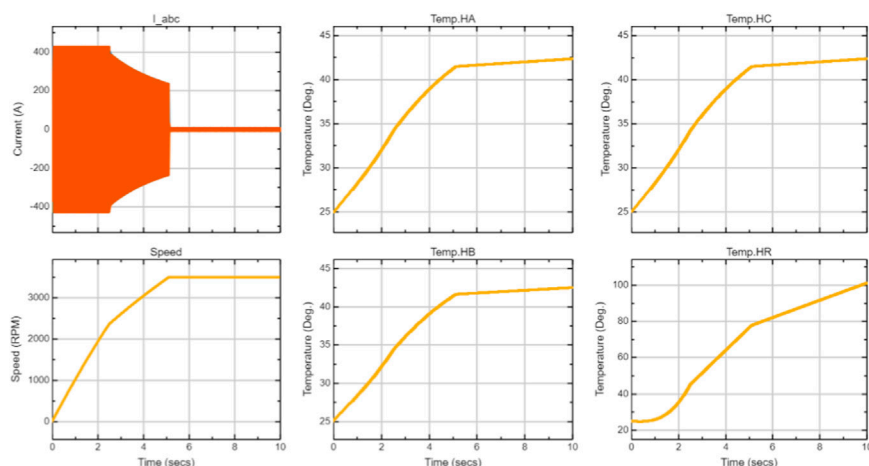


FIGURE 5
PMSM current, speed, and thermal port temperatures under constant load torque of 10 Nm for 10 s.

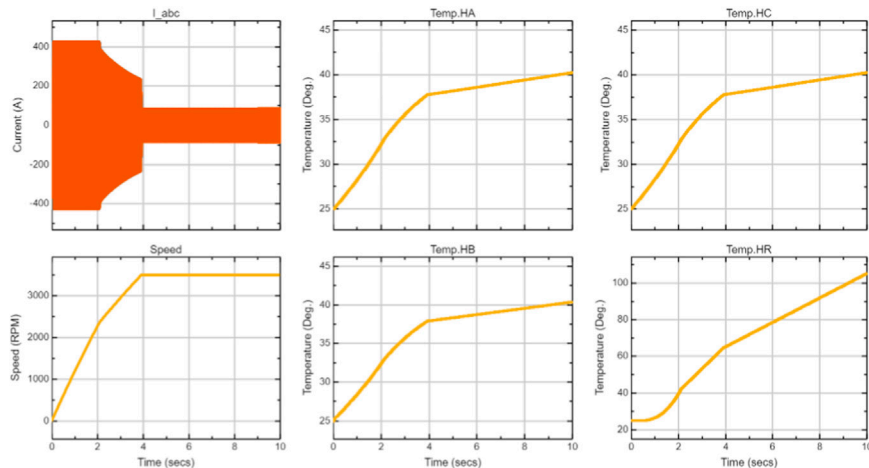


FIGURE 6
PMSM current, speed, and thermal port temperatures under constant load torque of 50 Nm for 10 s.

proportionally, and the temperatures of the stator (HA, HB, HC) and rotor (HR) rose continuously. The temperature growth rates accelerated significantly after 30 s of simulation time, corresponding to the high-load region (100–120 Nm). This cumulative heating effect is consistent with the thermal build-up observed in PMSM drive experiments by Ranjan *et al.* (Fu *et al.*, 2025), where extended high-torque operation caused exponential temperature to rise in rotor cores due to limited internal heat dissipation. Importantly, both stator and rotor temperatures exceeded their defined safe thresholds, confirming that passive thermal dynamics alone are insufficient for sustainable EV operation.

The simulation showed that between 40 s and 50 s, when the motor operated at 120 Nm, the temperature limit was compromised, leading to thermal runaway, a sharp uncontrolled increase in temperature primarily in the rotor winding (HR). This behaviour mirrors findings in El-Refaie (El-Refaie, 2013), who emphasized that

rotor magnet regions are most vulnerable to thermal saturation and demagnetization during overload conditions.

The results highlight the necessity of a staged cooling strategy for practical EV applications. Without active cooling or torque derating, continued operation beyond 50 s would result in irreversible thermal stress. Hence, incorporating a multi-stage proportional cooling mechanism (as modelled in Section 3.4.2) becomes essential to maintain the PMSM within its safe temperature limits under dynamically varying loads.

4.4 Staged cooling effectiveness

Figure 11 presents the temperature control response of the PMSM when the staged cooling strategy was activated following the temperature threshold exceedance observed in the previous variable-torque simulation (Figure 10). The staged cooling model

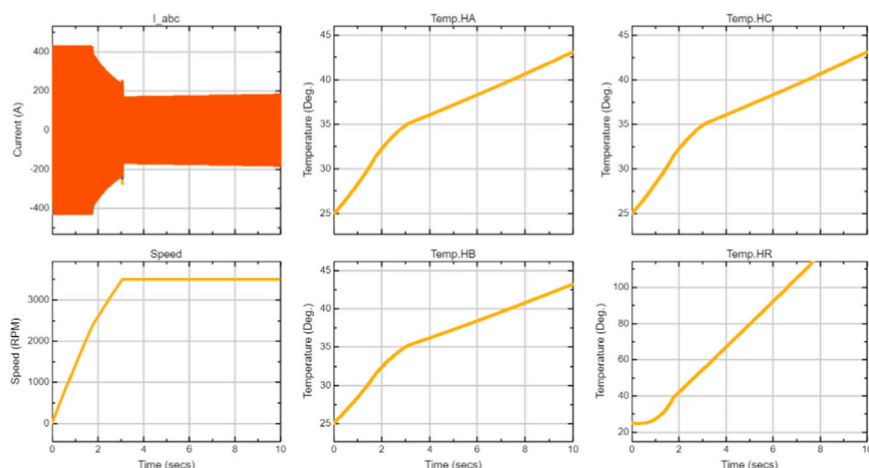


FIGURE 7
PMSM current, speed, and thermal port temperatures under constant load torque of 100 Nm for 10 s.

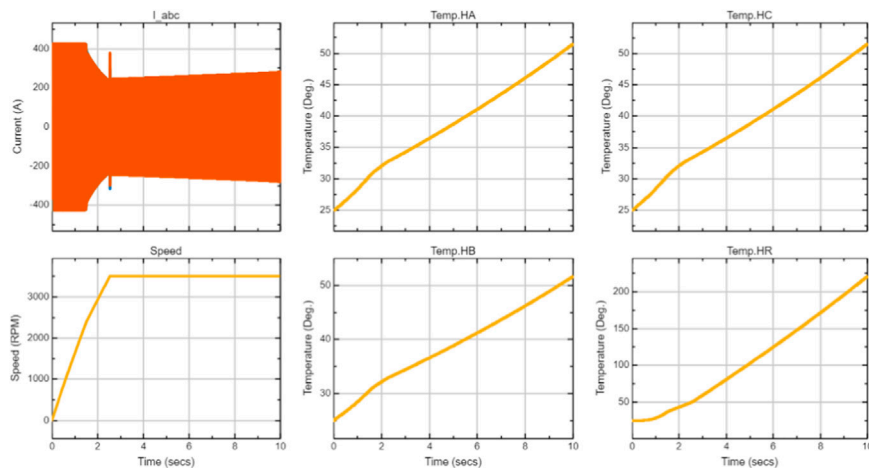


FIGURE 8
PMSM current, speed, and thermal port temperatures under constant load torque of 150 Nm for 10 s.

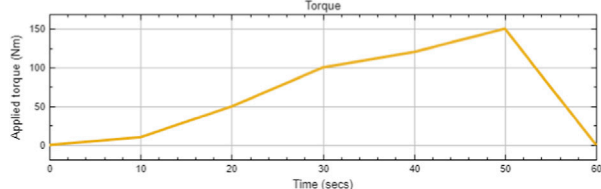


FIGURE 9
Applied torque profile for variable load simulation.

employed three proportional cooling levels, triggered sequentially as the winding or rotor port temperatures approached defined limits:

- i. Stage 1: activated at 80 °C (natural convection),
- ii. Stage 2: activated at 100 °C (forced air cooling),

- iii. Stage 3: activated at 115 °C (liquid cooling).

This logic mimics an intelligent cooling system in real EV drives, where multiple fan or coolant pump speeds are activated in response to temperature escalation to prevent component degradation and maintain continuous operation. Upon activation of the first cooling stage, the temperature rise rate in the stator windings (HA, HB, and HC) noticeably decreased, stabilizing the system near 95 °C–100 °C. As the rotor temperature (HR) continued to climb beyond 100 °C due to its higher thermal inertia, the second cooling stage engaged automatically, improving heat dissipation and reducing the rotor's temperature gradient. When HR approached 115 °C, the third and final cooling stage was activated, resulting in a clear temperature plateau for all ports, as seen in Figure 11. The staged cooling successfully suppressed the onset of thermal runaway that occurred in Figure 10 and restored thermal equilibrium across the system. The rotor temperature (HR), which initially rose faster and higher than the stator ports, was brought under control after

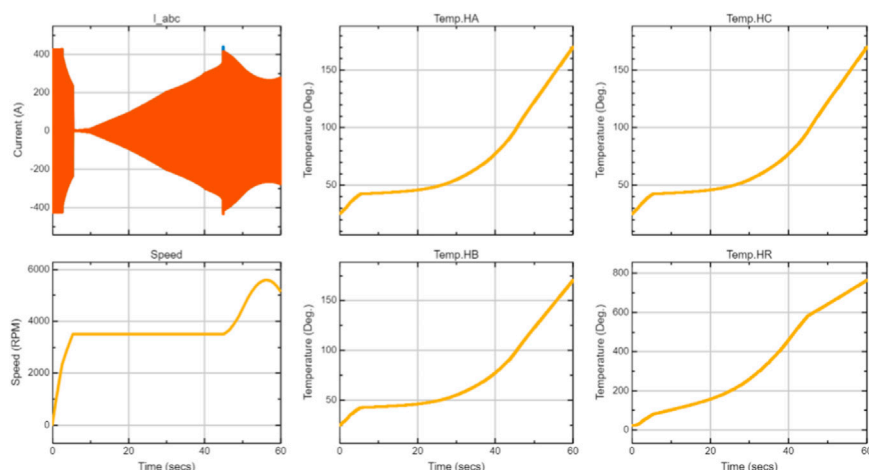


FIGURE 10
PMSM current, speed, and thermal port temperatures under sequential load torques: 0 Nm (0–10 s), 10 Nm (10–20 s), 50 Nm (20–30 s), 100 Nm (30–40 s), 120 Nm (40–50 s), and 0 Nm (50–60 s).

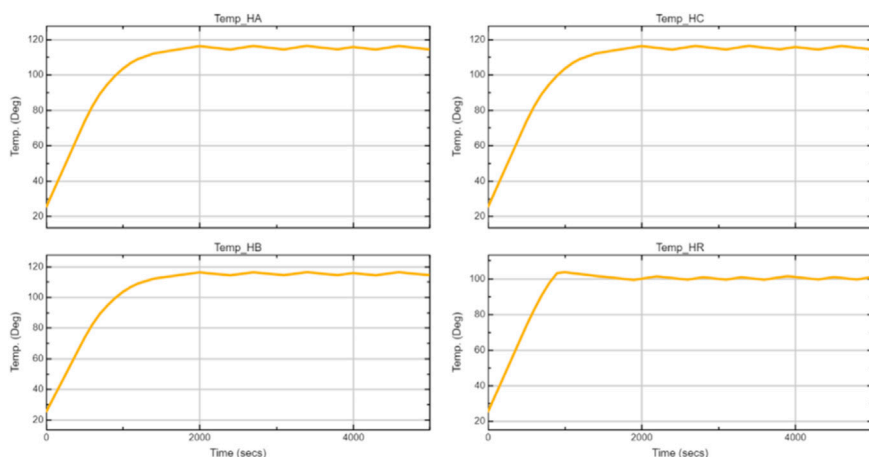


FIGURE 11
Controlled temperature rise in HA, HB, HC, and HR using staged cooling to prevent thermal runaway.

full cooling activation, with a slight temperature overshoot before steady-state recovery. Such behaviour is consistent with real-world observations of PMSM drives, where adaptive cooling delays but does not entirely eliminate transient temperature overshoots under peak torque demand.

This outcome validates the integration of the electro-thermal model with temperature feedback and staged cooling logic, demonstrating that even under continuous high-torque conditions, the PMSM can remain operational without invoking emergency shutdown. Similar multi-stage or variable-speed fan systems have been shown to reduce cumulative thermal stress and to lower battery power consumption which will improve the overall driving range in comparable PMSM studies (Nil Patel et al., 2022). Overall, the staged cooling approach implemented here represents a realistic and computationally efficient strategy for maintaining temperature stability in EV powertrains, effectively balancing dynamic performance and thermal safety. By moderating temperature rise

at each thermal port, the system not only prevents irreversible demagnetization but also preserves steady torque generation and avoids derating, which are crucial for reliable long-distance EV operation.

4.5 Comparison with literature

To validate the reliability and performance of the developed PMSM electro-thermal model, the simulated results were compared with established studies addressing field-oriented control (FOC), dynamic response, and thermal management of PMSM drives in electric vehicles. The comparison emphasizes three aspects: speed regulation, current and torque response, and thermal performance with cooling control. The proposed dual-loop control system demonstrated fast speed convergence and cruise tracking across all load conditions. Unlike most prior models, such as those presented by Zhou et al., (Zhou et al., 2020), which reported

TABLE 4 Performance metrics.

Metric	Proposed model	Published works
Settling time	~0.4 s	1.65 s using hybrid GA-RBL FOPID (Rajesh and Sebasthirani, 2025); 0.0847 s using Self-tuned FOPID (Yaried et al., 2025)
Max op. Temp	<120 °C (with staged cooling)	Evaluated up to 120 °C for baseline torque capacity; water-cooling reductions to 331.46 K (~58.3 °C) (Puma-Benavides et al., 2024)

gradual acceleration and limited torque response under heavy loads, the present model achieved faster convergence to the 3,500 rpm reference speed at higher torque values. This improvement stems from the adaptive torque-command scaling within the outer-loop controller, designed to prioritize speed tracking accuracy over efficiency during acceleration.

The current response exhibited smooth transients and minimal oscillation. The high start-up current decayed rapidly to a stable steady-state value once the target speed was reached, confirming effective decoupling of i_d and i_q components under FOC. This pattern aligns with the electromechanical characteristics reported by Dahl (2023), who demonstrated similar current stabilization behaviour in high-fidelity PMSM thermal models.

From a thermal perspective, the developed model closely replicated the temperature coupling between stator and rotor regions. Consistent with Zhou et al. (2020), and Jeon et al. (2023), the rotor (HR) exhibited a faster and higher temperature rise than the stator windings (HA, HB, and HC), due to internal iron losses and restricted convection within the rotor core.

During variable-load simulation, both stator and rotor exceeded their temperature thresholds when subjected to 120 Nm between 40 s and 50 s, resulting in thermal runaway. This confirms findings from König et al. (2023) that highlight the inadequacy of passive cooling under sustained high-load conditions.

By implementing a three-stage proportional cooling strategy, the proposed system effectively suppressed temperature overshoot and restored thermal equilibrium without shutdown. Similar multi-stage or optimized cooling control architectures have been demonstrated to extend safe operating duration and maintain torque capability in studies such as Cooling Strategy Optimization by Jeon et al. (2023). The results therefore validate the practical importance of staged thermal management for continuous EV operation.

To further contextualize the performance of the proposed dual-loop control and thermal management framework, the key simulation metrics under variable load conditions were benchmarked against recently published data for comparable EV traction drives, as summarized in Table 4.

While some recent studies report faster settling times and lower peak temperatures, these differences are largely attributable to motor capacity, physical inertia, and the modelled operating environments. For instance, the ultra-fast settling times (e.g., < 0.1 s) frequently cited in control optimization literature are typically

achieved using low-inertia motors or unloaded scenarios to highlight algorithm speed (Tripathi et al., 2023; Bayhan and Koçak, 2025; Lazi et al., 2025). In contrast, the PMSM modelled in this study is a 35 kW vehicle-grade traction motor with a substantial rotor inertia of 0.1243 kg·m². Achieving a 0.4 s settling time under heavy, real-world step loads (up to 150 Nm) demonstrates an optimal balance between rapid dynamic tracking and realistic electromechanical constraints.

Furthermore, the stabilized maximum temperature of 120 °C is highly appropriate for this power class. Benchmark studies demonstrating steady-state temperatures below 60 °C often model motors in highly constrained, thermally advantageous environments, such as underwater vehicles with infinite heat sinks, or utilize complex dual-channel liquid cooling (Tian et al., 2024; Chen et al., 2024; Zhou et al., 2025). For a terrestrial EV powertrain subjected to continuous 120 Nm to 150 Nm loads, a 120 °C ceiling remains well within the permissible limits of standard Class F (155 °C) and Class H (180 °C) insulation systems as defined by IEC 60085 (International Electrotechnical Commission, 2007). Additionally, motor thermal limits and allowable temperature rises specified in NEMA MG 1 (NEMA, 2016) further support the suitability of this operating range. This corresponds to approximately 77% and 67% utilization of Class F and Class H thermal limits, respectively. Maintaining operation below these thermal class limits significantly reduces insulation aging and degradation rates, thereby extending the service life of the traction drive. Therefore, the proposed staged proportional cooling strategy achieves thermal safety without the parasitic energy penalties or structural complexity associated with oversized liquid-cooling systems, validating its practical applicability for EV powertrain design.

5 Conclusion

This study developed and validated a comprehensive electro-thermal model of a permanent-magnet synchronous motor (PMSM)-driven electric-vehicle (EV) powertrain equipped with a dual-loop control strategy. The approach combined an outer-loop speed controller and an inner-loop current controller within a Simscape-based field-oriented control (FOC) environment, enabling detailed evaluation of both dynamic performance and thermal behaviour under varying load conditions. Thermal ports (HA, HB, HC, HR) were fully activated and linked with conduction and convection networks to emulate realistic heat transfer between stator, rotor, and ambient surroundings. Simulation results confirmed that the control strategy achieved fast and stable speed tracking across all torque levels while maintaining low torque ripple and minimal steady-state error. An interesting outcome of the controller tuning was that higher load torque shortened the acceleration period, demonstrating the priority of speed-tracking accuracy over efficiency during transients. Thermally, the model captured strong coupling between electromagnetic loading and heat generation, with the rotor core (HR) consistently experiencing a faster and higher temperature rise than the stator windings (HA - HC). When subjected to sequential load variations up to 120 Nm, the motor exceeded its thermal threshold, leading to potential runaway, highlighting the necessity of active temperature management. The implementation of a three-stage proportional cooling strategy

successfully prevented thermal failure by progressively activating cooling capacity at 80 °C, 100 °C, and 115 °C. This method effectively stabilized the system below 120 °C without interrupting torque delivery, outperforming single-threshold or on/off cooling methods reported in literature. Overall, the proposed model offers a validated and computationally efficient framework for analysing PMSM traction drives where electrical, mechanical, and thermal interactions must be concurrently evaluated. The integration of thermal feedback into the control loop enhances simulation realism and provides a basis for optimizing controller gains, cooling design, and fault-protection schemes in future EV development.

Further research will extend this framework to include:

- i. Inverter and battery thermal coupling, to evaluate complete drivetrain energy losses;
- ii. Real-time hardware-in-the-loop (HIL) validation, to verify controller stability under practical sensor and switching delays; and
- iii. AI-assisted adaptive cooling or torque-derating algorithms, to predict and mitigate thermal stress dynamically.

These extensions will strengthen the applicability of the model for advanced EV powertrain design and control optimization.

Data availability statement

The original contributions presented in the study are included in the article/supplementary material, further inquiries can be directed to the corresponding authors.

Author contributions

TS: Validation, Conceptualization, Writing – review and editing, Methodology, Software, Formal Analysis, Writing – original draft. OL: Funding acquisition, Project administration, Writing – review and editing, Validation, Supervision.

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